

1.

Q1 A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop. 10 marks

A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.

Write a Java program to:

Create an ArrayList of student names.

Read 8 student names from the user.

Traverse the list using an Iterator.

Display each student name on a separate line.

Display the total number of students stored in the list.

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         ArrayList<String> clg=new ArrayList<>();
6         int count=0;
7         for(int i=0;i<=7;i++){
8             clg.add(sc.nextLine());
9         }
10        Iterator<String> it=clg.iterator();
11        while(it.hasNext()){
12            System.out.println(it.next());
13            count++;
14        }
15        System.out.println("there are "+count+" number of students");
16        sc.close();
17    }
18 }
```

Run

Save

Input

ria
raj
kiran
sonu

Output

ria
raj
kiran
sonu
junaid
rohit

2.



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```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc=new Scanner(System.in);
5         ArrayList<Integer>a=new ArrayList<>();
6         for(int i=0;i<=9;i++){
7             a.add(sc.nextInt());
8         }
9         Iterator<Integer>it=a.iterator();
10        while(it.hasNext()){
11            Integer c=it.next();
12            if(c%2==0){
13                it.remove();
14            }
15            else{
16                System.out.println(c);
17            }
18        }
19        sc.close();
20    }
21 }
```

Output

Status: **Accepted**

Memory Used: 12096 byte | Execution Time: 0.177 sec

99
33
55
91
37

Input

12
32
54
66
99
33
55
98
91
37

3.

Q3 Product Price Analysis Using Iterator

10 marks

A supermarket stores the prices of various products in an `ArrayList<Double>`. The manager wants to analyze the prices using an Iterator.

Write a Java program to:

Create an `ArrayList<Double>`.

Read the prices of 10 products from the user.

Traverse the list using an Iterator.

Calculate and display:

Total price of all products.

Average product price.

Highest product price.

Display all product prices using the Iterator.

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int i;
6         double tot=0.0;
7         ArrayList<Double> a=new ArrayList<>();
8         for(i=0;i<=9;i++){
9             a.add(sc.nextDouble());
10        }
11        Iterator<Double> it=a.iterator();
12        while(it.hasNext()){
13            double cha=it.next();
14            tot+=cha;
15            System.out.println(cha);
16        }
17        a.sort(null);
18        //int lar=a[-1];
19        System.out.println("Total price is "+tot);
20        System.out.println("Average="+tot/10.0);
21        System.out.println("Highest product is:"+a.get(a.size()-1));
22
23    }
24 }
```

Input

```
23.54
11.43
87.40
89.98
```

Output

```
89.98
Total price is 617.7
Average=61.77
Highest product is:98.76
```